AI-use disclosure

Which AI tools were used, what they produced, and what was checked independently

1 AI tools and records

The existing `AI_USAGE_LOG.md` records AI-assisted formulation, traffic-model and baseline code, metrics, QUBO derivation, calibration scripts, and report tables for **Blocks 1-3**. It does **not name the AI product or model** used in those blocks. **Codex** assisted with the **Block 5-8** implementation and release drafting in this task thread.

Python, Qiskit, Qiskit Aer, SciPy, pytest, ReportLab, and Poppler were used for execution, testing, PDF generation, and visual checks; these are software dependencies/tools, not claims about the historical AI model.

2 AI-generated work

AI assistance produced formulation ideas, model and baseline code, QUBO construction and derivation text, calibration and metric scripts, the QAOA circuit/optimizer and artifact writer, Block 6 and Block 7 analysis scripts, automated tests, and the current report/disclosure/README drafts.

> The original `AI_USAGE_LOG.md` stops at Block 3; the later work is evidenced in the current source tree and task outputs, not in that original log.

3 Independent checks recorded in AI_USAGE_LOG.md

The log records the author's check of the four official R vectors (**116.0, 146.0, 237.0, 92.5**), review of scenario parameters against the problem statement, review of baseline and elite-set logic, and checking the derivation equations against `src/qubo.py`. It does **not** record the author's independent sign-off on all-256 scorer parity, QUBO/Ising parity, or circuit/count consistency. The repository contains automated tests for those checks; an automated pass is not represented as a personal verification.

4 Author confirmation required before submission

Leave each box unchecked until you have personally confirmed the statement.

- ☐ I independently confirm `scorer.py` parity on all 256 schedules.
  The test suite compares R and 100 generated scenarios; the current AI log records the four published vectors only.
  Author initials: _____ Date: _____
- ☐ I independently confirm QUBO-to-Ising energy equality, including sign and offset, for all 256 assignments.
  Author initials: _____ Date: _____
- ☐ I independently confirm the saved bound circuit, parameters, and 4,096-shot counts are consistent and reproducible.
  Author initials: _____ Date: _____
- ☐ I independently confirm I have read `docs/derivation.md` and understand its approximations and limits.
  The AI log records an equation review.
  Author initials: _____ Date: _____

> **To complete before submission:** historical AI product/model names are missing from `AI_USAGE_LOG.md`. Add them if known; do not infer them from this release task.